FLORENTIN NIYONGERE

Software Engineer | Autonomous AI Systems | Systems Engineering

florentinniyongere.fn@gmail.com • (320) 282-8788 • linkedin.com/in/niyongere • github.com/Umufundi

PROFILE

Software engineer building AI-enabled workflows and verifiable systems for regulated operations. Experienced in Python, API orchestration, data integrity, deterministic controls, and independent replay-based evaluation. interested in empirical AI safety and agent reliability.

TECHNICAL STRENGTHS

Python, SQL, JavaScript/TypeScript, C# • FastAPI, React, Node.js, PostgreSQL, Redis, MongoDB • AWS, Docker, REST APIs, EDI 270/271 • PyTorch, TensorFlow, scikit-learn, LangChain, LLM APIs • Workflow orchestration, verification, evaluation

PROFESSIONAL EXPERIENCE

FY Network, Inc. | Co-Founder & Software Engineer | Jan 2026–Present

- Co-founded FY Network to develop autonomous operations software for specialized industries. defined a reusable five-stage workflow - purpose, assembly, execution, approval, outcome - combining data, rules, people, and existing tools.
- Engineering Carelendr's staffing and compliance orchestration, representing caregiver availability, qualifications, authorizations, and service constraints as structured state for deterministic candidate ranking and placement.
- Developing a synthetic care-network simulation and independent verification harness with lifecycle state machines, authorization ledgers, scenario-driven failure cases, and replayable results. documented boundaries between tested components and incomplete end-to-end loops.
- Established cross-domain product direction spanning Carelendr (care staffing and compliance), Ref&Go (consent-based reference checks), and Rendus (personal financial coordination). distinguishing product demonstrations from independently verified deployment.
- Implemented and reviewed scoped access and simulation-worker security boundaries, including database role restrictions and isolation of synthetic provider data from real authorization records.

3A Homecare Agency | Software Engineer, Automation Systems | May 2024–Present

- Built a production authorization pipeline using Power Automate, Dropbox, PDF extraction, and Ankota APIs. Automated intake and verified writes while routing uncertain documents to staff for review.
- Connected Medicaid eligibility, client creation, and service goals to daily Microsoft Teams cards. Gave colleagues actionable visibility into eligibility changes, failed writes, and cases requiring decisions.
- Formalized cumulative Medicaid service variance per authorization, reconciling closed visits against authorized weeks using shared temporal cutoffs, 15-minute billing-unit quantization, gapped service periods, and PA-specific identity boundaries.
- Found a reload identity defect linked to ~4.1M orphaned authorization-week rows (98.7%). redesigned ingestion around stable PA numbers and atomic PostgreSQL loads to preserve referential integrity.
- Independently rebuilt the reconciliation logic from raw SDR JSON and Ankota data. matched 120/120 fields in an initial replay and 16/16 PAs in a second replay, reducing build time from ~20 minutes to 8.4 seconds (~143×).
- Replaced visit-file exports with a read-only Ankota SQL/API delta sync. matched visits by client, procedure, authorization week, and service range, excluding gap periods and preserving closed-visit-only accounting.
- Built case-manager-triggered visit approval workflows with API-based triage. Preserved human sign-off so case managers could review their own caseloads rather than manually reconstructing records.
- Developed a six-factor explanation of authorization variance (gaps, canceled/unclosed visits, over/short delivery and rounding), reconciled exactly on seven test PAs.
- Built a separate production Day Hab application that turns client ISP/SDR goals into accessible voice and text questions. DSPs draft and edit AI-assisted notes while managers retain final review authority.

- Empowered DSPs and managers with delegated daily setup, group assignments, note review, and exports. Automated weekday overdue-note emails and 7 PM Teams escalation cards that identify missing work and the oldest pending items.

GoAgentic | Software Engineer, Agentic Outreach | Jul 2022–Apr 2024

- Architected distributed lead qualification (Python, FastAPI, PostgreSQL) processing 10K+ daily interactions with ML models for regional business patterns.
- Built multilingual NLP pipeline using LangChain and fine-tuned Mistral/Llama models for Swahili, English, and French and reported 90%+ intent-classification accuracy.
- Developed event-driven WhatsApp integration with Redis streams and LLM agents to maintain contextual conversations and reported 70% faster responses.
- Implemented real-time lead analytics with Kafka, TimescaleDB, and Grafana, plus a ChromaDB RAG knowledge system for region-specific response relevance.
- Designed regional knowledge retrieval with ChromaDB and custom embeddings for context-aware customer outreach.

EARLIER ENGINEERING AND PRODUCT EXPERIENCE

Yahoo | Product Manager Intern | Jun–Aug 2021

- Analyzed 1M+ advertising impressions across carousel, native, and video formats. Developed ad-optimization roadmap that secured a $100K testing budget, following stakeholder analysis.
- Partnered with engineering and science teams on experiments, A/B testing, product requirements, and measurement of engagement and conversion.

Medtronic | Software Engineer Intern | May–Aug 2020

- Developed C#/.NET, SQL, and healthcare application services. Implemented verification, unit, and integration tests for Medtronic Application Services.
- Improved data retrieval and user-facing search workflows. Investigated defects and validated changes in a regulated healthcare environment.

Bloomberg | Black in Tech Fellow / Accelerator Participant | 2022

- Participated in engineering workshops, Jupyter-based portfolio exercises, and a six-week technical accelerator.

RESEARCH EXPOSURE AND LEADERSHIP

Montreal AI Ethics Institute | Remote AI Ethics Opportunity | May–Jul 2019

Participated in a remote opportunity focused on AI ethics.

NorthCloud / NSBE | Technical Community Leadership | 2020–2022

Founded student technical career-development initiatives. Organized engineering interview preparation, mentorship, and conference access. Served as NSBE chapter president.

EDUCATION

St. Cloud State University | Computer Science - Undergraduate Studies | 2017–2022

Relevant Coursework: Data Structures and Algorithms, Design and Analysis of Algorithms, Database Theory and Design, Introduction to Artificial Intelligence, Object-Oriented Software Development.

SELECTED RECOGNITION

Code2040 Fellow (2019–2020) • Microsoft Scholarship (2019, 2020) • NSBE Google Hackathon First Prize (2019)